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Visualizing technical analysis☆

Ruidong Yi, Jiajie Wang *

Shenzhen Campus, Jinan University, Shenzhen, Guangdong, 518074, China

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ABSTRACT

We devise an innovative technical analysis strategy that leverages combinations of technical indicators and machine learning techniques. Our models use image representations generated by visualizing the time-series data of individual stocks over a specific period to identify price patterns. To avoid data snooping, we carefully curtail the size of our training sets and shuffle the sample sets between models. Our approach yields highly reliable and impactful predictions on the U.S. stock market from January 1992 to December 2022 compared to multiple benchmarks, including momentum and reversal, traditional technical trading rules, industrial portfolios, and the Fama–French five-factor model. Using the GradCAM method, we visualize the attention maps of model predictions, providing strong interpretability to our approach of technical analysis.

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1. Introduction

Many researchers and practitioners are drawn to the challenge and potential rewards of predicting stock trends. One popular approach is technical analysis, which focuses solely on historical data, including prices and volumes. While the efficient market hypothesis suggests that technical analysis cannot generate significant profits in an efficient market, it continues to attract attention (see Lo & Hasanhodzic, 2010; Lo, Mamaysky, & Wang, 2000; Schwager, 2012). Additionally, behavioral economics theory proposes that emotions and market practitioners' biases can impact prices (see Barberis, 2018), allowing for the study of historical trends. Momentum and reversal effects further suggest the possibility of identifying price patterns and achieving profits (see Schwert, 2003).

Still, it is challenging to create or choose the appropriate technical indicator, due to the need to adjust numerous parameters and the lengthy process this entails. To deal with this issue, machine learning (ML) methods have been introduced in technical analysis that not only assist investors in selecting and constructing indicators but also broaden the horizons of technical analysis. In the investment sector, an assortment of ML techniques are used to identify patterns in price trends. However, ML techniques have certain challenges, such as data snooping and a lack of interpretability.

Moreover, an extensive array of academic articles have been published, employing ML approaches and diverse features to facilitate portfolio management and asset allocation. Among these, Nazareth and Reddy (2023) provide a systematic examination of the integration of ML in finance, pinpointing six key areas: stock markets, portfolio management, cryptocurrency, foreign exchange markets, financial crises, and bankruptcy and insolvency. Analyzing 126 articles, Nazareth and Reddy (2023) found that ML, particularly long short-term memory models, excelled at stock market prediction. Similarly, Pinelis and Ruppert (2022) harnessed random forest ML to refine portfolio

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* Corresponding author.
E-mail address: wang_jj@sz.jnu.edu.cn (J. Wang).

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allocation by focusing on predicting returns and volatility, outperforming traditional strategies and offering a cohesive approach to balancing investment risk and return. Ma et al. (2021) explored the application of ML and deep learning for stock return prediction and portfolio optimization, emphasizing the efficacy of various models at enhancing mean–variance and omega portfolio strategies. They concluded that the random forest-based mean–variance model excelled, especially when considering transaction costs.

Xiu et al. (2020) furthered the discourse by conducting a comparative analysis of various ML methods within the realm of empirical asset pricing, particularly for assessing asset risk premiums. The study underscored the significant economic gains for investors using ML forecasts, doubling the performance of top regression-based strategies in some instances. The standout methods were trees and neural networks, which excelled due to their ability to capture nonlinear predictor interactions. They emphasized the importance of momentum, liquidity, and volatility as predominant predictive signals, setting a new benchmark for the predictive accuracy of ML in asset pricing.

Both Xiu et al. (2020) and Giglio et al. (2022) shed light on the challenges inherent in asset pricing, such as the intricacy of measuring expected returns due to market efficiency and the overwhelming dimensionality of predictor variables relative to sample size. They also addressed concerns like structural breaks, regime switches, and nonstationarity, which complicate analysis. Within this context, both articles positioned ML as a transformative force in empirical asset pricing, equipped to tackle these challenges through sophisticated variable selection, dimension reduction, and other advanced techniques.

In addition to demonstrating ML's potential in fundamental analysis, many studies have been conducted to evaluate the profitability of ML-based technical analysis. Teixeira and De Oliveira (2010) pioneered this integration by proposing a method that unifies the k-nearest neighbors classifier with foundational technical analysis tools like stop loss, stop gain, and the relative strength index (RSI). Their rigorous performance evaluation against a buy-and-hold strategy revealed that their innovative approach could substantially amplify profits for the majority of companies, concurrently reducing market exposure by minimizing buy operations.

Building upon this, Dash and Dash (2016) advanced the field by integrating a computationally efficient functional link artificial neural network (CEFLANN) with a suite of technical analysis rules. This CEFLANN was adept at deciphering complex nonlinear relationships between key technical indicators, yielding trading signals that were instrumental in tracking market trends, and executing trades based on predefined criteria. This hybrid approach was shown to surpass other ML models, including support vector machines, naive Bayesian models, k-nearest neighbors, and decision tree models, in terms of profitability.

Gerlein et al. (2016) extended the validation of simple ML models into the financial trading sphere, specifically within the dynamic foreign exchange market. By selecting attributes such as the RSI and Williams %R, along with other price-related features, they concluded that these simple ML models could be effectively harnessed for financial trading, achieving profitability that rivaled more complex systems.

Peng et al. (2021) took a deeper dive into the application of deep neural networks and feature selection techniques for stock price direction forecasting, utilizing a comprehensive set of 124 technical analysis indicators. Their study, which spanned daily stock data across seven global market indexes from 2008 to 2019, applied various feature selection methods and neural network architectures to predict stock price movements. The findings provided valuable insights into the predictive capabilities of different indicators, the enhancement of model performance through feature selection, and the economic benefits of the derived strategies, all while considering transaction costs and market efficiency.

In a related vein, Ayala et al. (2021) explored the synergy between ML techniques and traditional technical analysis indicators to refine stock market prediction strategies. They proposed a hybrid approach that merges a technical indicator with ML methods to generate trading signals, with a spotlight on the triple exponential moving average and the moving average convergence/divergence (MACD). They evaluated the performance of ML models such as linear models, artificial neural networks, random forests, and support vector regression on daily trading data from three major indices: Ibex35 (IBEX), DAX, and the Dow Jones Industrial (DJI). The results indicated that the integration of ML techniques can significantly bolster the effectiveness of traditional technical analysis, leading to more robust trading signals and enhanced competitiveness in the financial markets.

Nevertheless, these studies viewed technical indicators as numerical inputs, ignoring technical trading rules (e.g. the golden cross and death cross) that are essentially based on visual features that contain a plethora of nonlinear features. Only a few studies have considered this issue. Cohen et al. (2020) proposed a technical analysis method based on image classification. They trained a neural network to identify sell or buy signals based on opening, high, low, and closing (OHLC) charts, Bollinger bands, and moving averages, and they created an automated model for chart marking, thereby reducing the workload of technical analysis. However, they did not explore the profitability of the ML-based strategy. To address this, Jiang et al. (2023) developed another technique based on image classification to determine whether visualization contributes to technical analysis. They used a simple conversion of numerical price data into images containing an OHLC chart, moving average line, and trading volume chart, which were then used to train a convolutional neural network (CNN) that learned the price pattern and could predict future price trends. On the one hand, this approach reported significant profits generated by the strategy. On the other hand, it placed great emphasis on the interpretation of the model. Their study significantly contributed to encouraging researchers to view price trends as images instead of numerical data, thereby continuing the exploration of the relationship between visualizing price trends and technical analysis.

However, the effectiveness of technical analysis based on ML should be scrutinized. Arnott et al. (2018) delved

into the application of ML in investment management, highlighting its promise and the risks of misapplication due to the limited data available in finance. The authors urged a cautious approach, emphasizing the necessity of selecting appropriate ML applications and developing a rigorous research protocol to minimize overfitting and data mining biases. They proposed a detailed seven-point protocol covering aspects from research motivation to model complexity and cross-validation, aiming to reduce errors and false discoveries in backtesting investment strategies. They concluded with a call for a research culture that prioritizes scientific integrity over immediate success, suggesting that such a culture, along with the adoption of a robust protocol, is vital for enhancing investment outcomes and maintaining the credibility of investment managers. Regarding technical analysis, data snooping is one of the most significant issues. According to Park and Irwin (2007), 'When data snooping occurs, any successful results may be spurious because they could be obtained just by chance.' Data snooping occurs when a dataset is used multiple times for model selection. One of the notorious characteristics of the neural network model is that one can always try out different model architectures to fit the data, sometimes even tweaking the random seed works, suggesting that data snooping is a common practice when employing a neural network model for technical analysis. One could devise an outperforming model by chance or simply select the best one from a huge model set. A similar issue was raised by Joubert et al. (2024), highlighting shortcomings of the most widely used backtesting method—walk-forward testing: 'One of the principal limitations of the approach is that only a single path is tested, which raises the risk of overfitting. …the walk-forward method, by definition, does not require practitioners to have a comprehensive understanding of the underlying processes or knowledge of the reasons for a strategy's success or failure.' Consequently, it is imperative that methodologies incorporating complex parameterization to derive exceptional strategies from historical data are underpinned by a solid theoretical framework and are interpretable in terms of their superior performance. This ensures that the strategies are not only historically effective but also have a clear and justifiable foundation, which is crucial for their potential to deliver consistent results in actual market conditions.

The false discovery rate is a well-known approach to identifying data snooping, as explored by Bajgrowicz and Scaillet (2012). According to their findings, none of the 7846 technical trading rules tested were significant once the false discovery rate was considered. The false discovery rate is a method designed to control false positives (i.e. Type I errors). However, this method may not be suitable for ML-based technical analysis. This is because various models in a model set can learn different information about the training data (i.e. convergence to different local optimums due to different initial conditions), making it crucial to determine whether a model uncovers genuine patterns or outperforms others due to luck. One way to determine whether the model possesses genuine predictability is to build a long-short portfolio that ranks cross-sectional stocks based on the signal generated by the model, a method commonly employed by empirical analysis to evaluate the profitability of factors like momentum and value (see Asness et al., 2013). This method was also employed by Jiang et al. (2023). However, the issue of using a dataset repeatedly for model selection remains unaddressed.

Our study focuses on exploring the potential benefits of visualization techniques in technical analysis and evaluating the economic feasibility of developing a reliable model for forecasting future price trends.

Our analysis includes various technical indicators: the MACD, RSI, multiple moving averages, and Bollinger bands. We also utilize colored OHLC charts overlaid with various technical indicators commonly used in traditional technical analysis.

We utilize the Swin transformer model proposed by Liu et al. (2021), one of the leading models in computer vision. Specifically, this model is embedded with self-attention to build interrelations between different chunks of the input image and to focus on the critical information, which helps the model better understand the OHLC charts and identify predictive patterns. We opt for the Swin transformer model for two main reasons. First, it is one of the best-performing models in the area of computer vision,[1] making it highly competitive for image classification tasks. Second, the self-attention mechanism is similar to how humans perceive images and enables us to visualize the attention map of the model, providing some interpretability to the model.

In this study, we take several measures to overcome the problem of data snooping. First, we divide our sample into 10 equal-length periods and randomly select four periods for training, one for validation, and five for testing. We train 12 model sets with different random seeds while keeping the other hyperparameters the same to assess the model's robustness. Second, we randomly select a small portion of the training sets to train different models within a model set for each training epoch. This not only improves the performance during validation but also reduces the chances of using the same dataset multiple times for model selection. Finally, like previous studies, we use different deciles of prediction scores to build a long-short portfolio that provides a comprehensive evaluation of the model's predictability.

Our key finding is that utilizing neural network strategies to visualize technical indicators can be highly profitable, even when accounting for transaction fees. Furthermore, our analysis demonstrates that these strategies outperform recent momentum and reversal strategies while providing steady, risk-adjusted returns across all test periods. Additionally, the models tend to perform better on sample periods further back in time, which we attribute to advancements in computational power and data accessibility. We also compare these strategies to traditional technical trading rules. The results show that the ML portfolios outperform all rules in all periods. Compared to industrial portfolios and the Fama–French five-factor model, ML strategies demonstrate significant annual abnormal returns. Moreover, a transaction cost sensitivity

[1] For a survey, see Guo et al. (2022).

analysis verifies the robustness of models under various market conditions. Lastly, to address interpretability, we use the GradCAM method to visualize the attention map in the form of a heat map and overlay it on the original image input. This shows that the models have learned the essentials of technical analysis and can utilize the embedded information within the technical indicators to make predictions.

Overall, our findings provide compelling evidence that supports the value of developing a technical analysis model that combines robust computer vision with visualized technical indicators. Our study contributes to the ongoing exploration of price patterns based on image representations.

2. Visualization of time-series data

While most literature employing ML methods and technical indicators relies on numerical data, our study pioneers a new frontier by visualizing time-series data. This innovative approach transforms the way we perceive and analyze stock trends. In this section, we explain how to convert time-series data into images. We begin by demonstrating the approach we use to convert data into image coordinates and then explain the rendering of the image representation of the OHLC chart and technical indicators.

2.1. Normalizing and rescaling the data

Our methodology utilizes a variety of technical indicators, including volume, multiple moving averages, the MACD, the RSI, and Bollinger bands. Moving averages and Bollinger bands are overlaid on the OHLC charts (see the example in Fig. 3) and are defined as main-chart indicators. The MACD, volume, and RSI are sub-chart indicators, because they have different domains from the OHLC data. The main-chart indicators are normalized and rescaled along with the OHLC data, whereas the sub-chart indicators are normalized and rescaled independently.

We use the x axis to represent time and the y axis to represent the prices or indices of indicators. Let D_i represent a set of time-series data coordinates of the indicator i:

$$\begin{aligned} D_i &= \{(v_1, t_1), \ldots, (v_n, t_n)\} \\ D_m &= D_{OHLC} \cup D_{i \in \text{main-chart indicators}} \\ D_s &= D_{OHLC} \cup D_{i \in \text{sub-chart indicators}} \\ P &= \{(x, y) | (x, y) \in N \times N, x \leq W, y \leq H\} \end{aligned}$$

where P denotes a pixel space, and W and H respectively denote the width and height of the pixel space. Each image has a pre-specified starting time, denoted as c, and a pre-specified time window, denoted as w. We use c and w to confine the range of the data rendered. We have

$$f_m : D_m \mapsto P, f_m(v, t; c, w) = (x, y)$$

where

$$y = H - \left\lfloor \left(\frac{3}{4}H - 2\right) \times \frac{v - \min\limits_{\substack{(v,t)\in D_m \\ t\in(c,c+w]}} v}{\max\limits_{\substack{(v,t)\in D_m \\ t\in(c,c+w]}} v - \min\limits_{\substack{(v,t)\in D_m \\ t\in(c,c+w]}} v} + \frac{1}{4}H + 2 \right\rfloor \tag{1}$$

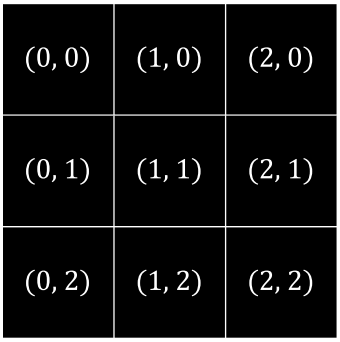


Fig. 1. Coordinates in the pixel space.

$$x = (t - c) \times 3 \tag{2}$$

and where H is pre-specified and W is determined by $W = 3 \times w$. As we use three pixels' width to represent each day's information, the total width of the picture is the length of the observing window multiplied by three.

First, Eq. (1) normalizes time-series data values to a range of $[0, 1]$. Then, it rescales them to a range of $[0, 3/4 \times H - 2]$, which sets the main chart's height to be $3/4 \times H - 2$. Subtracting two pixels creates a margin between the main chart and sub-chart. The term $1/4 \times H +$ 2 moves the main chart to the image's top. The floor operation ensures non-negative integer values. In the pixel space, coordinates with smaller y values appear above those with greater y values (refer to the illustration in Fig. 1). The additional operations outside the floor brackets flip the y coordinates to account for this property.

We assume that each day spans three pixels. Eq. (2) ensures that the x coordinates align with the middle pixel of each day.

The sub-chart indicators occupy the bottom quarter of the image's height, and we have

$$f_s : D_s \mapsto P, f_s(v, t; c, w) = (x, y)$$

where:

$$y = H - \left\lfloor \left(\frac{1}{4}H - 2\right) \times \frac{v - \min\limits_{\substack{(v,t)\in D_s \\ t\in(c,c+w]}} v}{\max\limits_{\substack{(v,t)\in D_s \\ t\in(c,c+w]}} v - \min\limits_{\substack{(v,t)\in D_s \\ t\in(c,c+w]}} v} \right\rfloor \tag{3}$$

$$x = (t - c) \times 3 \tag{4}$$

2.2. OHLC chart

Opening, high, low, and closing prices are contained in the raw data and are used to generate the OHLC chart. We render the OHLC chart in candlestick style, which can be defined as follows. The OHLC bar consists of a rectangle and a vertical line in the middle for a given day; the rectangle represents the opening and closing prices, whereas the vertical line represents the high and low prices.

In this study, we consider that if the opening price is higher than the closing price, the OHLC bar will be

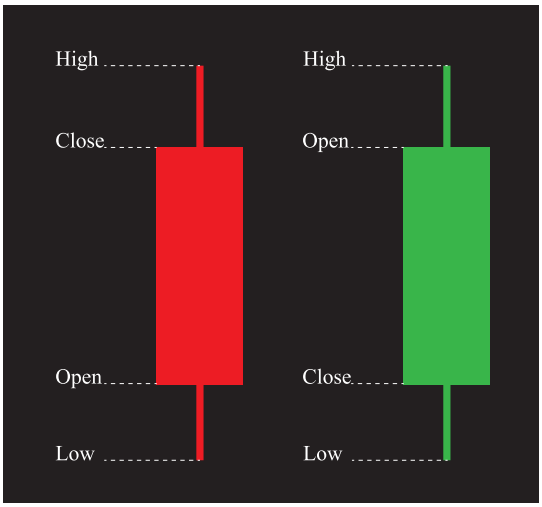


Fig. 2. Candlestick-style OHLC bar.

rendered green, and if the opening price is not higher than the closing price, the OHLC bar will be rendered red. Fig. 2 provides an example of an OHLC bar. Multiple OHLC bars when combined form an OHLC chart, and each OHLC bar is three pixels wide in our image representation.

2.3. Technical indicators

Examples of our image representations of different technical indicators are shown in Fig. 3.

2.3.1. Moving averages

Traders utilize numerous moving averages to analyze past price trends on different time scales. The data are then used to forecast future price movements. According to research by Zhu and Zhou (2009), moving averages can surpass model-based optimal trading strategies, even when the stock price is uncertain. Our paper uses three moving averages: the five-day, 20-day, and 60-day, which are rendered yellow, blue, and orange, respectively, in our image representation.

When the closing price traverses the moving average curves from a lower position to a higher one, it is interpreted as a buy signal. Conversely, when the closing price crosses the moving average curves from above to below, it signifies a sell signal. Additionally, the occurrence of a moving average with a shorter time period crossing above (below) a moving average with a longer time period can also serve as a buy (sell) signal.

Moving averages are useful when determining the general price trend. However, they are insensitive to recent abrupt price changes and may generate lagged signals. Thus, traders often combine them with other indicators.

2.3.2. Bollinger bands

Bollinger bands, proposed by Bollinger (2002), sharpen the sensitivity of moving averages to reflect a market's volatility more precisely. The indicator generates three results, the upper, middle, and lower bands, which are calculated as follows:

$$middle_t = \frac{1}{n}\sum_{i=0}^{n-1} close_{t-i}$$

$$upper_t = middle_t + a\sqrt{\frac{1}{n}\sum_{i=0}^{n-1}(close_{t-i} - middle_t)^2}$$

$$lower_t = middle_t - a\sqrt{\frac{1}{n}\sum_{i=0}^{n-1}(close_{t-i} - middle_t)^2}$$

where n and a are parameters. Bollinger bands essentially draw borders around n-day moving averages at a distance of $a \times \sigma_{\text{n-days}}$. Hence, Bollinger bands serve as a mechanism for generating trade signals akin to moving averages. However, they nullify these signals when the price action traverses the upper or lower boundaries of the bands. In this study, we set them to 10 and 2, and the lower, middle, and upper bands are rendered purple, cyan, and yellow, respectively.

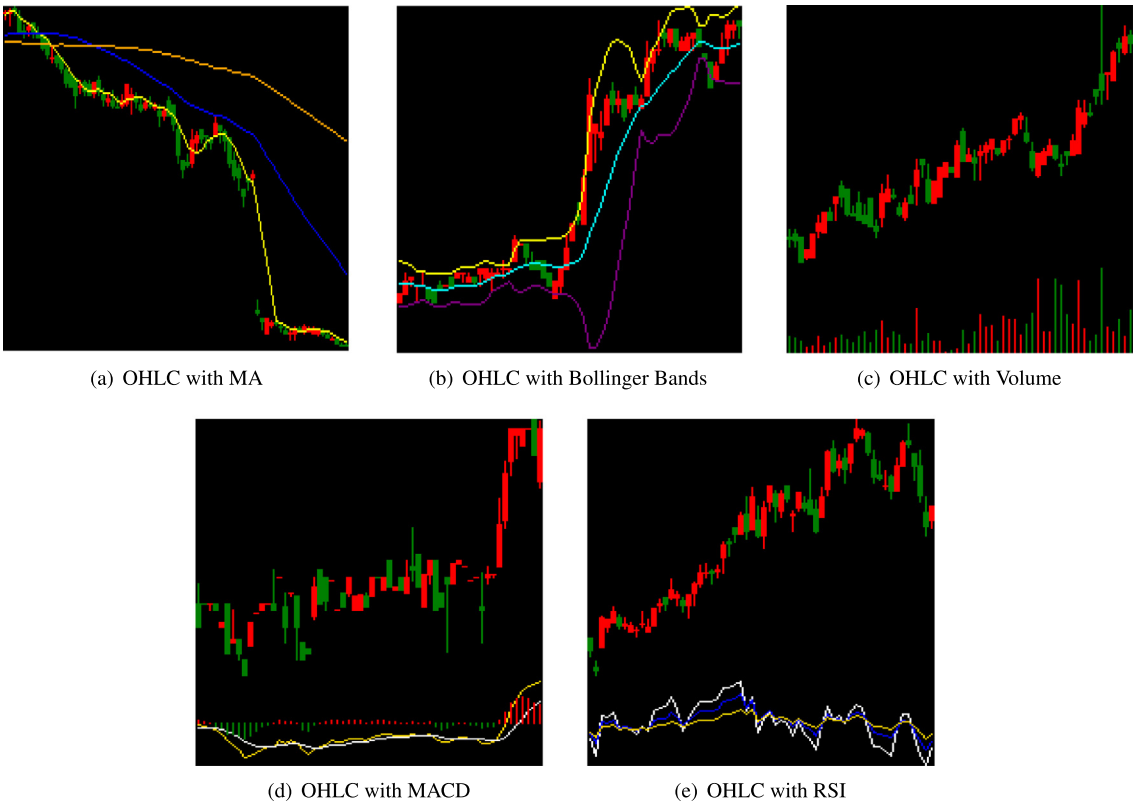
2.3.3. Volume

Blume et al. (1994) found that technical analysis may be more applicable to some small-cap stocks with low attention, stating that in highly volatile market conditions, volume can provide valuable information for traders. Dupont (1997) analyzed the different information possessed by traders and the distribution characteristics of their trading volume and price within the framework of static rational expectations, and found that if traders had more consistent expectations, the mean trading volume and price variance would decrease, and otherwise increase. This indicates that the volume–price relationship of stocks can, to some extent, determine the level of consensus among traders on stock prices, which may enable us to predict future price trends.

In our study, we render the volume for each day as a vertical bar with a width of one pixel, and each bar has the same x coordinate as the corresponding OHLC bar's vertical center line. The color of the volume bar is aligned with the OHLC bar.

2.3.4. MACD

The MACD indicator is a widely used technical tool among traders (see Appel & Dobson, 2007; Murphy, 1999). It measures the relationship between two moving averages to identify changes in trends and momentum, providing a clear analysis of market conditions to make informed trading decisions. Chong and Ng (2008) conducted a study on the FT30 index showed that MACD strategies perform better than buy-and-hold strategies. In a follow-up study (see Chong et al., 2014), they tested different parameter settings on various indicators in different markets. The results showed that MACD(12,26,0) was profitable in the Italian and Canadian markets. We further explore the profitability of utilizing MACD with ML methodologies.



(a) OHLC with MA (b) OHLC with Bollinger Bands (c) OHLC with Volume

(d) OHLC with MACD (e) OHLC with RSI

Fig. 3. Visualization examples of different technical indicators with OHLC charts.

To calculate the MACD indicator, it is first necessary to acknowledge the exponential moving average (EMA):

$$EMA(n, x_t) = \begin{cases} \frac{n-1}{n+1} EMA(n, x_{t-1}) + \frac{2}{n+1} x_t & t > 0 \\ x_t & t = 0 \end{cases}$$

where n is the parameter of the EMA, determining the smoothness of the output array.

The MACD indicator consists of three components: the MACD curve, signal curve, and histogram.

$$\begin{aligned} MACD_t &= EMA(n, close_t) - EMA(m, close_t) \\ signal_t &= EMA(p, MACD_t) \\ histogram_t &= MACD_t - signal_t \end{aligned}$$

where $n, m (n < m)$ and p are parameters.

The MACD is essentially the difference between a faster EMA curve and a slower EMA curve. The signal is the EMA of the MACD. The histogram is the difference between the MACD and signal. A buy signal is activated when the MACD curve ascends from beneath the signal curve to a position above it, and conversely, a sell signal is generated when the MACD curve descends from above the signal curve to a position below it. Additionally, this is equivalent to the histogram transitioning from negative values to positive values for a buy signal, and from positive values to negative values for a sell signal. In practice, the drawback of the MACD indicator is that its signals often lag behind the market and it sometimes generates false signals when the price oscillates frequently within a range.

In this study, the parameters for this indicator are set to 12, 26, and 9. We draw the MACD curve in gold, the signal curve in white, and the histogram in red and green. The zero axis of the histogram is placed in the horizontal center line of the sub-chart. The histogram bar's color is red when it is above the zero axis; otherwise it is green.

2.3.5. RSI

The RSI is an indicator that measures the strength of a security by comparing its recent gains and losses. It is a momentum oscillator that helps traders identify the overbought and oversold conditions of the asset and provides signals to buy or sell (see Murphy, 1999). In the aforementioned studies (Chong & Ng, 2008; Chong et al., 2014), RSI strategies performed better than buy-and-hold strategies on the FT30 index. The RSI (21, 50) performed exceptionally well in the Milan Comit General Index and the S&P/TSX Composite Index, while the RSI (14, 30/70) performed well on the Dow Jones index. In a more recent study conducted by Gumparthi (2017), the authors investigated the validity of the RSI in Indian stock markets and concluded that it could be effectively used to construct portfolios and successfully predict the future trends of the stocks in most cases. It is intriguing to test how this indicator performs with ML techniques.

The RSI is calculated as follows:

$$up_t = \begin{cases} close_t - close_{t-1} & close_t > close_{t-1} \\ 0 & close_t \le close_{t-1} \end{cases}$$

$$down_t = \begin{cases} close_{t-1} - close_t & close_t < close_{t-1} \\ 0 & close_t \ge close_{t-1} \end{cases}$$

$$sup_t = \frac{n-1}{n} sup_{t-1} + \frac{1}{n} up_t$$

$$sdown_t = \frac{n-1}{n} sdown_{t-1} + \frac{1}{n} down_t$$

$$RSI_t = 100 - \frac{100}{1 + \frac{sup_t}{sdown_t}}$$

where n is the parameter controlling the smoothness of the curve.

The greater (smaller) that n is, the blunter (sharper) the curve is. Traders often utilize different RSI curves and observe their patterns as a whole. Typically, when the RSI curve with a shorter period rises and intersects the RSI curves with longer periods, this occurrence is interpreted as a buy signal. Conversely, when the RSI curve with the shorter period descends and crosses below the RSI curves with longer periods, it is regarded as a sell signal. We employ three RSI curves with n parameters of five, 10, and 20, which are rendered in white, blue, and gold, respectively.

While the RSI's ability to adapt quickly to market fluctuations is a notable advantage, its pattern is largely random, presenting a challenge for traders seeking to forecast trends and make decisions.

3. Data and model

With our visualization techniques established, we now delve into the empirical backbone of our study: the dataset and the model architecture. Understanding these elements is crucial for grasping how our technical analysis strategy is executed. In this section, we first explain the details of our dataset and then we demonstrate the hyperparameter settings and the framework of our model training. Finally, we briefly introduce the architecture of the Swin transformer and how its features facilitate our specific task.

3.1. Data

Our dataset contains daily data for almost all firms listed on the New York Stock Exchange (NYSE) and the NASDAQ,[2] ranging from June 1992 to December 2022. The variables include OHLC and volume. Observations with at least one empty value are abandoned because the missing value may distort technical indicators calculated based on the raw data and create gaps in the OHLC chart.

We separate our dataset into 10 equal-length periods, shuffle them, and then randomly pick five for testing, four for training, and one for validation. Different model sets have different separations of the training, validation, and test periods. Thus, we have 12 years for training, three years for validation, and 15 years for testing.

This approach of separating samples is similar to n-fold cross-validation in the field of ML. The main difference is that n-fold cross-validation requires that each period be used as a validation period at least once, and the test periods are fixed, while our approach shuffles not only training periods and validation periods but also test periods. Firstly, by shuffling test periods, we not only prevent model selection bias in the training process but also avoid selection bias in the test process. Secondly, if we exhaust all combinations of training, validation, and test periods, the magnitude of computation will be unaffordable. Thus, our choice of shuffling sample periods reduces the computation burden and tackles the problem of data snooping by giving each period an equal probability of being in the training samples as well as the test samples.

We choose three observing windows to generate images: five-day, 20-day, and 60-day. The return five days after the window determines the label for the image generated (0 for non-positive returns and 1 for positive ones).

In Section 2, we introduce the technical indicators: main-chart indicators (MA and Bollinger bands) and sub-chart indicators (volume, MACD, and RSI). This gives us a total of six combinations of technical indicators. We observe these indicators across three different windows, giving us a total of 18 combinations. We create a model set with 18 models, each one based on a different combination of technical indicators and observing windows (Table 1).

In order to account for the impact of randomness on the separation of samples and model training, we developed 12 sets of models, each comprising 18 models. All model sets are characterized by the same hyperparameters and setting of models within them, except for the random seed. Consequently, we trained a total of 216 distinct models throughout our testing process.

Table 1
Eighteen models in a model set.

No.	Window	Main-chart	Sub-chart
1	5 days	MA	Vol
2			RSI
3			MACD
4		BOLL	Vol
5			RSI
6			MACD
7	20 days	MA	Vol
8			RSI
9			MACD
10		BOLL	Vol
11			RSI
12			MACD
13	60 days	MA	Vol
14			RSI
15			MACD
16		BOLL	Vol
17			RSI
18			MACD

[2] The data come from CRSP, and 15 firms are excepted for a corrupted cumulative factor to adjust price (CFACPR) in their historical records.

3.2. Model training

In this section, we discuss the framework of our training process, the hyperparameter settings of our training process, and the issues we encountered during the training.

The training process is conducted on a model set basis. Sample periods are re-shuffled every time that the process of training, validating, and testing a model set is completed. For any given model, the loss function is the cross-entropy loss function. The optimization algorithm is AdamW (see Loshchilov & Hutter, 2017), with a learning rate of 10^{-8} and a weight decay rate of 10^{-8}. The batch size is 10, and the training iteration during every epoch is 50.[3] As for the early stopping and initialization procedures, we follow the setting of Jiang et al. (2023). An early stop is executed if the average epoch loss is not improved for two consecutive epochs, and the initialization method of model weights is Xavier Normal (see Glorot & Bengio, 2010). For every epoch, once the training iteration is completed, the validation process is then conducted to record the accuracy of the model's prediction on the validating periods. When all models in the model set complete their training, outperforming models are determined based on their average validating accuracy during the training. We set a threshold of 51% accuracy. Subsequently, the outperforming models are sent to the testing procedure.

We intentionally set a relatively small training iteration using a batch size of 10 and an iteration of 50. That is, during each epoch, only 500 images are being fed to the model, while the total number of available images in the training set is at a magnitude of 10^8. This is because, under an iteration greater than 10,000, almost all models converge to a locally optimal solution that makes the same prediction for all inputs. Although we tried different combinations of hyperparameters and replaced our basic model with models based on long short-term memory, CNNs, and MLPs, we failed to tackle this issue. A plausible explanation for this issue is that models cannot find any predictive pattern for these images. However, after we reduced the magnitude of the training iteration, some models could output meaningful predictions. This is counter to the well-accepted knowledge that neural networks learn better when they consume more samples. Our conjecture is that the sample set, as a whole, is hard to learn because not all samples are predictable based on the information presented in the image. Meanwhile, during a specific window, the technical indicators may not generate any meaningful signal, thus containing nothing to learn from. These samples could undermine the ability of the model to learn from these predictive patterns, and they account for a large portion of the sample set because the model forgets what it has learned from the small number of learnable samples and chooses to overfit those meaningless samples. We call these samples with no valuable information 'toxic samples'. The fewer the number of images used in the training process, the fewer the number of toxic samples fed to the model. In that case, the model is more likely to learn predictive patterns.

After several rounds of trial and error, we found that a training iteration of 50 was appropriate. This setting not only helped us resolve the issue mentioned above but also coincidentally addressed the problem of data snooping because the probabilities of overlapping training samples between models were small.

3.3. A brief introduction to the swin transformer

We chose the Swin transformer (T version) as our basic model for training.

The transformer, first proposed by Vaswani et al. (2017), is a sequence transduction model for translation tasks. It is the first of its kind that is solely based on the attention mechanism. Subsequently, the transformer was utilized in computer vision by Dosovitskiy et al. (2020) and reached an outstanding speed–accuracy tradeoff on image classification compared to CNN. Liu et al. (2021) further explored the potential of the transformer in the area of computer vision and proposed the Swin transformer, which achieves an even better speed–accuracy tradeoff on image classification and dense recognition tasks.

The most important idea of all transformers is the attention mechanism, which can be expressed as

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

where $softmax(\cdot)$ is a function that standardizes n-dimension vectors[4]; and Q and K are $d_{model} \times d_k$ matrices.

In Eq. (5), the variables Q, K, and V represent matrices of row vectors. Specifically, q and k stand for queries and keys, respectively, while v represents values. The attention mechanism matches a key to each query and outputs a corresponding value. The term QK^T computes dot products of the row vectors in Q and K, where $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$. Greater dot products between two vectors indicate higher similarity. To prevent dot products from becoming excessively large, a scaling factor of $1/\sqrt{d_k}$ is applied. The *softmax* function calculates the score for each match and multiplying by V yields the weighted sum of all values. The score determines the weight assigned to its corresponding value.

While there are numerous variants of the attention mechanism, they start by embedding an input into a numerical space (e.g. word-to-vector in natural language processing). Then, the vector, denoted by x, is projected into another space by parameter matrices W^Q, W^K, and W^V.[5] The main difference lies in how inputs are converted into vectors and how Q, K, and V are computed.

[3] The learning rate, weight decay rate, batch size, and iteration times were decided with manual fine-tuning using tensorboard, a training visualization tool provided by PyTorch. We conducted several short runs to determine the appropriate hyperparameters, considering the convergence of training loss and whether the model degenerated to make the same predictions during training (a locally optimal solution).

[4] Suppose x is an n-dimension vector, $softmax(x_i) = \frac{x_i}{\sum_{j=1}^{n} x_j}$. When applied to a matrix X, $softmax(x_{i,j}) = \frac{x_{i,j}}{\sum_{k=1}^{n} x_{i,k}}$.

[5] $q = x_q W^Q$, $k = x_k W^K$, and $v = x_v W^V$.

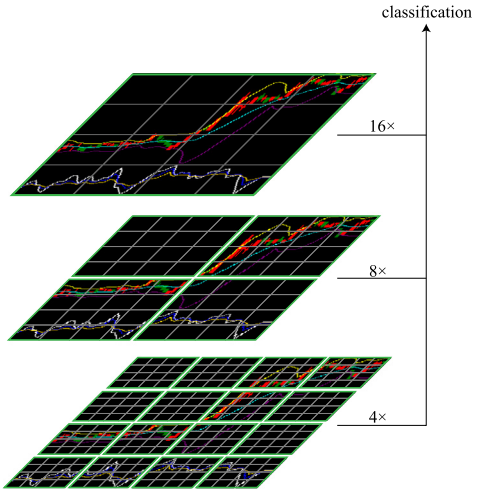


Fig. 4. Hierarchical feature maps built by the Swin transformer. Numbers are the downsampling multipliers for different stages. Chunks colored green are windows, while gray chunks are patches.

The Swin transformer employs two variants of the attention mechanism: the window-based multi-head self-attention (W-MSA) and shifted window-based multi-head self-attention (SW-MSA).

The term 'multi-head' indicates that a given embedding vector, x, is cut into n (n stands for the number of heads) equal-length pieces, and Q_i, K_i, and V_i are computed from each piece by head i with matrices W_i^Q, W_i^K, and W_i^V. Furthermore, the attentions calculated by heads are concatenated together and multiplied by a parameter matrix W^O to output the final attention. The term 'self-' means that Q, K, and V are calculated from the same input x. The terms 'window-based' and 'shifted window-based' are explained below, after we illustrate the hierarchical feature maps built by the Swin transformer.

The Swin transformer understands an image by building hierarchical feature maps of the image. In the first stage, the image is divided into 16 windows, each of which contains 16 patches. The computation of attention is executed on patches and confined to each window. Different windows cannot communicate with each other within a specific stage. In the next stage, every four neighbors of windows form a new window, cut into 16 new patches, and the attention within each new window is recalculated. This process, called downsampling, is repeated until no new windows can be formed. Interrelationships of different parts of the image are determined through the process of downsampling. During this process, the model perceives the image with decreasing details and increasing field of sensing. This process is illustrated in Fig. 4. Thus, the term 'window-based' refers to the partitioning of windows and calculating attention separately based on these windows.

The term 'shifted window' pertains to an additional operation that is taken between every two stages to calculate the attention. Its purpose is to better model the connections between adjacent windows. Fig. 5 illustrates the process of window shifting.

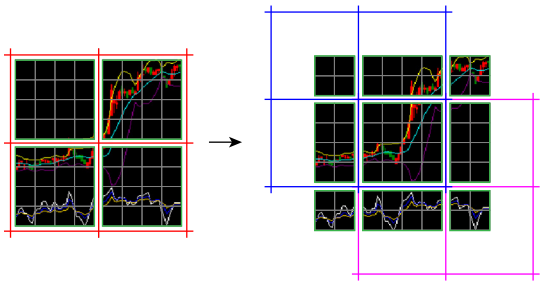


Fig. 5. Shifted window. The grid (shown in red) used to partition windows is shifted to north-west (shown in blue) and south-east (shown in purple), re-cutting new windows that connect the original ones.

The way it processes images means that the input image fed to the model has a size of 224 × 224. Thus, the H parameter in Eqs. (1) and (3) is set to 224. We only rescale the width of images when feeding them to models.

A survey by Arkin et al. (2023) states a key architectural difference between the Swin transformer and CNN as follows: CNNs extract high-level abstract features using convolutional kernels, but their effective receptive field is often smaller than the theoretical one, which hinders the full use of contextual information. In contrast, transformers excel at capturing global contextual information and establishing long-range dependencies between objects, enabling the extraction of more useful features.

In our scenario, the superiority of this approach lies in its enhanced capability to model OHLC charts compared to CNNs. Transformers are adept at identifying correlations between distant segments of the chart, a feature that CNNs may neglect due to their convolutional processing. Specifically, in instances where the current price is breaking a previous resistance level, the model must be capable of recognizing and establishing long-range dependencies between the most recent price data and those from a more remote timeframe to fully interpret this market movement.

There are several reasons we prefer the Swin transformer to a CNN. First, as mentioned above, it performs better than a model based on a CNN with parameters of similar magnitude. Second, the idea of partitioning an image into windows and assigning different weights of attention to them is analogous to the way humans comprehend an image; that is, you first pay attention to different levels of details of an image and then put them together. Third, the attention maps are available during the model inference, thus allowing us to better understand the model's predictions. Lastly, the architectural features of the Swin transformer are superior for building long-range dependencies compared to a CNN.

4. Empirical results

With these architectural benefits in mind, we now turn to our empirical results to evaluate the Swin transformer's performance in practice. In this section, we

focus on the evaluation of the trained models. We begin by demonstrating the overall performance of long-short portfolios built based on model predictions and then thoroughly investigate the portfolio's performance on different test periods and deciles. Additionally, we utilize multiple momentum strategies, reversal strategies, and strategies based on traditional technical indicators as benchmarks to compare the performance. Furthermore, we employ Fama–French five-factor models and 10 industry portfolios to gain a comprehensive understanding of the performance of our models. Meanwhile, we evaluate the models' sensitivity to transaction fees under different market volatility environments. These exercises enable us to determine the efficacy of the trained models and the economic feasibility of implementing such models. Finally, we use the GradCAM approach to visualize attention maps generated during model inference.

4.1. Overall performance

We use a rolling window to make predictions on a daily basis for each stock. The outputs of our models are scores indicating the price trend for the next five days of the inputs. Every stock is assigned a score on a given day. We first rank them cross-sectionally from high to low to rescale the magnitude and standardize each day's cross-sectional scores:

$$\widetilde{S}_{i,t} = rank(S_{i,t}) \tag{6}$$

$$w_{i,t} = \frac{\widetilde{S}_{i,t} - median(\widetilde{S}_t)}{5\sum_j^N |\widetilde{S}_{j,t}|} \tag{7}$$

where $w_{i,t}$ is the weight of total funds we invest into stock i on day t. Our funds are evenly invested every five days, so we add a multiplier of five in the denominator.

Among the 216 models trained, only 22 surpassed the designated threshold and were consequently included in the testing procedure. A summary of the overall performance of these 22 models during the test periods is presented in Table 2. Notably, only three models—20BOLL+MACD, 5BOLL+RSI, and 5MA+Vol—yielded returns that are statistically significant at a level of less than 5%. However, it is worth highlighting that only the 5MA+Vol and 5BOLL+RSI models generated meaningful risk-adjusted returns.

4.2. Benchmark

4.2.1. Comparison with momentum and reversal strategies

Our further test focuses on the three aforementioned models. We build benchmark strategies for different models based on their model sets' test periods.

We define our momentum strategy and reversal strategies in the following forms:

$$MOM_{i,t}(n) = \frac{close_{i,t} - close_{i,t-n}}{close_{i,t-n}} \tag{8}$$

$$REVS_{i,t}(n) = \frac{\frac{1}{n}\sum_{\tau=0}^{n-1} close_{t-\tau} - close_t}{\frac{1}{n}\sum_{\tau=0}^{n-1} close_{t-\tau}} \tag{9}$$

Table 2
Overall performance of long-short portfolios of models on test periods. Returns and standard deviations are annualized. ***, **, and * stand for returns and Sharpe ratios significant at 1%, 5%, and 10% significance levels, respectively. The risk-free rate is the average yield of three-month U.S. Treasury Bills for all test periods. The transaction fee is assumed to be 0.1‰.

Model set	Model name	ret	stddev	SR
1	20BOLL+RSI	−0.86%	0.75%	−4.98
	20MA+RSI	−1.58%	0.63%	−7.06
	20MA+Vol	*2.55%	1.37%	−0.24
	60BOLL+MACD	−1.18%	0.98%	−4.13
	60MA+RSI	0.75%	0.99%	−2.15
6	5MA+Vol	−5.06%	1.37%	−5.65
	5MA+RSI	−4.75%	1.33%	−5.61
	5BOLL+Vol	−3.62%	0.92%	−6.84
	5BOLL+RSI	−3.53%	0.89%	−7.02
	20MA+Vol	−2.81%	0.86%	−6.39
9	60MA+Vol	−0.29%	0.66%	−4.27
	60BOLL+Vol	−0.30%	0.65%	−4.35
12	5MA+RSI	−3.55%	1.45%	−4.18
	5MA+Vol	***7.79%	1.87%	***2.82
	60BOLL+RSI	−0.83%	0.90%	−3.69
	60MA+Vol	−0.28%	1.07%	−2.6
	5BOLL+Vol	−4.42%	1.57%	−4.41
	5BOLL+RSI	***5.66%	1.51%	**2.08
	5BOLL+MACD	1.54%	1.21%	−0.8
	20MA+MACD	−5.16%	1.91%	−4.02
	20BOLL+MACD	**4.60%	1.80%	1.16
	5MA+MACD	1.83%	1.46%	−0.47

Table 3
Break-even transaction costs of models on different market volatility periods. The higher the break-even costs, the more robust the model is during the corresponding market volatility period with regard to transaction costs.

	20BOLL+MACD	5BOLL+RSI	5MA+Vol
Extreme	0.00117	0.00102	0.00143
High	0.00038	0.00092	0.00179
Moderate	0.00036	0.00046	0.00046
Low	0.00126	0.00185	0.00175

We use the parameters five, 20, and 60 for momentum and reversal strategies in responding to the different window lengths of neural network models. The re-balance period for these portfolios is five days, and funds are equally invested every five days, the same as the portfolios of neural networks. The results are shown in Table 4.

Based on the table provided, it is evident that all four neural network strategies delivered better results during the 1990s and generated noteworthy returns with respect to risk, while the mom5 strategy was the only one capable of competing with them. As for the most recent periods (2019–2023), the risk-adjusted returns for all strategies declined.

We consider the following explanation for this phenomenon. Dugast and Foucault (2023) studied the correlation between data abundance and data miner performance. Their findings suggest that data abundance could eventually impede the performance of data miners. In the early years, traders were less informed about technical analysis and ignorant of price patterns due to insufficient data and computational power, which left abundant profit opportunities for utilizing such techniques. As

Table 4
Portfolio performance comparison in Model Set 12. The risk-free rate is the average yield of three-month U.S. Treasury Bills for the specific test period. The transaction fee is assumed to be 0.1‰.

Panel A. Test period 1: from 1992-01-01 to 1995-02-06

	swin20BOLL+MACD		swin5BOLL+RSI		swin5MA+Vol		mom5		mom20		mom60		revs5		revs20		revs60	
decile	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR
1	−1.17%	−1.21	−5.70%	−1.38	−1.73%	−0.42	−21.82%	−4.02	−15.69%	−3.18	−12.29%	−2.55	−6.83%	−0.94	***22.33%	***3.26	**10.73%	**1.98
2	−4.22%	−1.00	0.07%	0.02	−3.82%	−0.89	−17.89%	−4.08	−11.11%	−2.52	−8.02%	−1.80	***11.43%	***2.59	5.81%	1.30	−3.33%	−0.76
3	−2.46%	−0.54	0.31%	0.07	−0.22%	−0.05	−10.55%	−2.47	−8.45%	−1.86	−4.47%	−1.00	0.32%	0.08	−2.22%	−0.52	−5.08%	−1.17
4	0.78%	0.17	4.19%	0.94	2.84%	0.63	−9.36%	−2.24	−5.44%	−1.23	−3.35%	−0.78	−1.47%	−0.35	−3.61%	−0.85	−2.94%	−0.71
5	3.90%	0.84	*8.19%	*1.82	6.34%	1.43	−4.36%	−0.97	−4.10%	−0.93	−6.44%	−1.46	−4.61%	−1.06	−1.98%	−0.46	−5.69%	−1.30
6	*9.26%	*1.91	**11.43%	**2.46	**9.27%	**2.00	−3.70%	−0.84	−4.59%	−1.03	−3.58%	−0.80	−5.79%	−1.32	−4.35%	−1.00	−5.13%	−1.19
7	*9.00%	*1.88	**10.84%	**2.47	**11.14%	**2.48	−3.13%	−0.72	−4.58%	−1.09	−4.40%	−1.05	−6.64%	−1.56	−5.78%	−1.39	−2.89%	−0.65
8	*9.13%	*1.84	***12.79%	***2.79	***14.12%	***3.07	−0.34%	−0.08	−3.80%	−0.89	−3.40%	−0.78	−12.78%	−3.05	−10.40%	−2.30	−6.04%	−1.37
9	***13.22%	***2.68	***14.99%	***3.35	***16.72%	***3.68	5.77%	1.27	1.43%	0.33	−4.91%	−1.12	−17.97%	−4.04	−15.61%	−3.50	−9.49%	−2.13
10	***17.77%	***3.64	***19.28%	***3.13	***21.75%	***3.41	***66.62%	***6.40	−17.33%	−1.36	−20.98%	−1.66	−27.62%	−3.17	−7.20%	−0.31	−18.03%	−2.11
H-L	***8.45%	***7.47	***14.39%	***5.15	***12.92%	***4.83	***47.15%	***7.52	−11.64%	−1.19	−21.03%	−1.84	−12.52%	−2.58	−3.84%	−0.20	−15.27%	−3.28

Panel B. Test period 2: from 1998-03-14 to 2001-04-19

	swin20BOLL+MACD		swin5BOLL+RSI		swin5MA+Vol		mom5		mom20		mom60		revs5		revs20		revs60	
decile	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR
1	4.71%	−0.06	−0.45%	−0.05	−6.74%	−0.88	−38.32%	−3.30	−25.02%	−2.10	−15.00%	−1.25	***70.83%	***4.92	***87.83%	***5.43	***50.96%	***4.00
2	0.66%	0.07	2.88%	0.33	−2.61%	−0.29	−15.60%	−1.59	−8.43%	−0.83	−5.30%	−0.52	*18.84%	*1.91	11.04%	1.13	7.82%	0.79
3	1.77%	0.19	4.90%	0.53	6.37%	0.65	−8.38%	−0.87	−3.28%	−0.34	−4.88%	−0.50	4.74%	0.50	4.96%	0.52	3.28%	0.35
4	5.65%	0.58	8.96%	0.93	10.69%	1.06	−4.55%	−0.48	−2.93%	−0.31	−0.66%	−0.07	2.26%	0.24	2.27%	0.24	1.79%	0.19
5	10.24%	1.02	15.08%	1.52	11.73%	1.17	−2.10%	−0.22	−2.89%	−0.31	−1.72%	−0.18	−0.49%	−0.05	0.98%	0.11	0.02%	0.00
6	13.01%	1.23	15.28%	1.50	15.94%	1.54	−1.07%	−0.12	−1.13%	−0.12	0.34%	0.04	−2.67%	−0.29	−1.09%	−0.12	−0.93%	−0.10
7	14.00%	1.29	*16.74%	*1.66	*19.08%	*1.86	0.11%	0.01	0.21%	0.02	0.10%	0.01	−5.23%	−0.55	−3.05%	−0.33	−5.01%	−0.54
8	*18.69%	*1.67	**20.27%	**2.00	**23.08%	**2.24	2.67%	0.28	4.22%	0.45	0.51%	0.05	−9.70%	−1.03	−8.09%	−0.84	−3.61%	−0.37
9	17.25%	1.49	**21.58%	**2.10	***27.03%	***2.66	13.05%	1.34	6.29%	0.64	5.47%	0.55	−19.28%	−1.96	−13.55%	−1.35	−9.05%	−0.88
10	**25.74%	**2.20	**24.85%	**2.39	***31.06%	***3.11	***101.15%	***4.53	3.90%	0.21	0.09%	0.00	−35.15%	−2.84	0.60%	0.02	−19.60%	−1.31
H-L	***8.87%	***2.80	***11.12%	***5.13	***17.70%	***7.44	***70.58%	***5.01	1.70%	0.12	−0.70%	−0.04	−40.59%	−7.09	12.24%	0.52	−28.52%	−2.51

Panel C. Test period 3: from 2001-04-19 to 2004-05-26

	swin20BOLL+MACD		swin5BOLL+RSI		swin5MA+Vol		mom5		mom20		mom60		revs5		revs20		revs60	
decile	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR
1	13.21%	1.44	13.16%	1.57	10.74%	1.35	−23.85%	−2.46	−10.77%	−1.17	0.07%	0.01	***80.92%	***7.05	***128.63%	***10.51	***107.43%	***9.80
2	13.49%	1.56	13.40%	1.55	10.88%	1.25	−4.20%	−0.50	9.60%	1.12	12.32%	1.43	***44.29%	***5.21	***34.83%	***4.11	***28.03%	***3.37
3	9.40%	1.03	14.48%	1.61	*15.20%	*1.67	7.08%	0.89	11.12%	1.37	12.22%	1.48	***26.03%	***3.21	***25.09%	***3.13	**18.72%	**2.30
4	10.43%	1.11	*16.07%	*1.75	14.17%	1.53	9.64%	1.25	12.93%	1.63	*13.75%	*1.72	**19.00%	**2.47	**19.98%	**2.58	**17.98%	**2.37
5	12.78%	1.33	**20.56%	**2.22	*16.71%	*1.75	10.17%	1.32	12.55%	1.58	**15.68%	**1.99	*14.63%	*1.93	*14.27%	*1.87	*13.97%	*1.85
6	14.19%	1.47	**20.59%	**2.20	**18.89%	**1.99	*13.86%	*1.83	**15.73%	**2.06	*14.22%	*1.85	11.80%	1.54	12.62%	1.62	*14.02%	*1.76
7	13.91%	1.41	**20.88%	**2.25	**21.79%	**2.29	**18.15%	**2.35	**18.15%	**2.36	**17.45%	**2.27	10.37%	1.33	12.65%	1.62	*14.12%	*1.77
8	*16.57%	*1.69	**21.49%	**2.33	***25.13%	***2.65	***23.45%	***2.95	**20.46%	**2.57	**18.98%	**2.35	5.31%	0.67	8.39%	1.04	10.77%	1.31
9	*18.16%	*1.86	**22.82%	**2.45	***27.15%	***2.90	***39.76%	***4.71	***27.69%	***3.27	***26.23%	***3.20	−4.08%	−0.49	2.63%	0.31	9.49%	1.10
10	**21.54%	**2.27	***28.20%	***3.02	***31.51%	***3.46	***133.42%	***8.17	**41.42%	**2.46	*49.68%	*1.73	−20.34%	−1.96	−8.21%	−0.52	−2.34%	−0.20
H-L	2.85%	1.49	***5.56%	***3.53	***8.15%	***4.14	***67.00%	***6.99	***32.18%	***2.63	41.87%	1.57	−34.88%	−5.94	−27.44%	−1.75	−25.61%	−3.19

Panel D. Test period 4: from 2007-07-02 to 2010-08-07

	swin20BOLL+MACD		swin5BOLL+RSI		swin5MA+Vol		mom5		mom20		mom60		revs5		revs20		revs60	
decile	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR
1	−6.37%	−0.65	−7.99%	−0.65	−10.37%	−0.83	−89.64%	−4.31	−81.24%	−4.71	−73.19%	−4.55	−95.01%	−2.99	−87.29%	−1.98	−61.99%	−3.23
2	−3.04%	−0.23	−5.27%	−0.40	−2.20%	−0.16	−56.92%	−4.40	−50.39%	−3.78	−49.80%	−3.76	−37.02%	−2.78	−39.38%	−2.95	−44.02%	−3.29
3	−0.55%	−0.04	−3.17%	−0.23	−1.30%	−0.09	−52.43%	−4.05	−52.51%	−3.97	−50.71%	−3.87	−42.69%	−3.29	−45.10%	−3.47	−48.50%	−3.66
4	1.13%	0.08	−1.45%	−0.10	−0.43%	−0.03	−53.72%	−4.17	−53.40%	−4.12	−53.95%	−4.06	−47.15%	−3.73	−44.95%	−3.45	−47.95%	−3.68
5	3.74%	0.26	0.88%	0.06	1.12%	0.08	−52.26%	−4.08	−51.26%	−3.92	−50.76%	−3.85	−47.13%	−3.68	−48.62%	−3.83	−49.43%	−3.90
6	5.12%	0.35	3.65%	0.26	1.72%	0.12	−50.27%	−3.94	−50.20%	−3.89	−50.72%	−4.02	−49.53%	−3.87	−49.11%	−3.83	−47.88%	−3.70
7	6.44%	0.44	2.58%	0.18	3.98%	0.28	−48.40%	−3.78	−49.28%	−3.85	−51.15%	−3.89	−51.05%	−3.94	−50.80%	−3.90	−50.44%	−3.93
8	8.25%	0.55	4.71%	0.33	5.59%	0.40	−45.13%	−3.47	−46.82%	−3.57	−46.61%	−3.54	−52.20%	−4.07	−50.55%	−3.91	−50.20%	−3.80
9	8.25%	0.56	5.49%	0.38	5.70%	0.42	−37.52%	−2.81	−43.53%	−3.27	−45.45%	−3.39	−55.11%	−4.32	−51.09%	−3.92	−48.91%	−3.74
10	10.75%	0.74	9.93%	0.70	6.97%	0.54	34.17%	1.40	20.18%	0.87	−3.24%	−0.13	−95.78%	−2.91	−97.60%	−1.33	−82.58%	−3.35
H-L	***8.08%	***3.12	***8.49%	***4.17	***8.39%	***3.51	***187.20%	***12.29	***73.56%	***3.86	17.53%	0.81	1.52%	0.10	−49.61%	−0.74	−27.29%	−1.80

Panel E. Test period 5: from 2019-11-25 to 2023-01-01

	swin20BOLL+MACD		swin5BOLL+RSI		swin5MA+Vol		mom5		mom20		mom60		revs5		revs20		revs60	
decile	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR
1	5.34%	0.45	3.68%	0.35	−2.32%	−0.24	−94.08%	−6.28	−88.51%	−6.31	−82.53%	−6.05	−98.67%	−4.42	−96.82%	−2.05	−87.49%	−4.41
2	5.21%	0.47	4.33%	0.39	0.42%	0.04	−62.40%	−5.37	−60.52%	−5.13	−57.62%	−4.67	−48.70%	−3.99	−52.10%	−4.26	−55.80%	−4.60
3	4.71%	0.41	6.06%	0.53	4.42%	0.38	−59.47%	−5.23	−58.62%	−5.12	−58.31%	−5.09	−52.42%	−4.49	−53.91%	−4.48	−54.96%	−4.62
4	5.32%	0.45	7.90%	0.67	6.43%	0.55	−56.78%	−5.02	−57.27%	−4.99	−56.06%	−4.93	−53.29%	−4.65	−52.84%	−4.63	−53.76%	−4.68
5	6.35%	0.53	8.11%	0.69	6.08%	0.51	−57.10%	−5.13	−58.23%	−5.25	−56.20%	−5.02	−53.32%	−4.89	−53.48%	−4.81	−52.75%	−4.74
6	7.71%	0.63	9.43%	0.80	10.11%	0.84	−58.89%	−5.24	−57.06%	−5.12	−58.50%	−5.39	−56.52%	−5.12	−55.37%	−4.99	−53.89%	−4.85
7	8.79%	0.71	10.95%	0.92	11.65%	0.95	−54.10%	−4.81	−55.52%	−4.98	−55.32%	−4.84	−56.75%	−5.05	−55.92%	−4.95	−55.89%	−4.93
8	7.46%	0.60	10.64%	0.89	14.31%	1.15	−53.42%	−4.52	−55.09%	−4.68	−56.13%	−4.79	−58.83%	−5.15	−58.22%	−5.08	−57.12%	−5.00
9	10.29%	0.82	12.15%	1.00	16.85%	1.34	−52.22%	−4.28	−54.18%	−4.47	−55.17%	−4.55	−63.17%	−5.47	−60.43%	−5.22	−58.51%	−4.90
10	12.85%	1.02	13.64%	1.11	*20.77%	*1.72	−97.98%	−1.28	−92.37%	−1.58	−96.25%	−1.50	−98.90%	−3.37	−99.77%	−1.08	−98.44%	−2.10
H-L	2.80%	1.10	*3.89%	*1.94	***10.37%	***3.53	−93.47%	−1.35	−81.71%	−1.59	−91.96%	−1.63	−13.61%	−0.71	−28.76%	−0.27	−22.43%	−0.45

more data and more robust computers became available, traders became increasingly aware of profitable patterns and exploited these opportunities.

Furthermore, we consider two bearish periods in U.S. market history: 2000–2003 and 2007–2008. Most of our models yielded statistically significant returns during these periods.

In Fig. 6, it can be seen that all of the neural network strategies generated a positive Sharpe ratio throughout all of the test periods, while the mom5 strategy exhibited outstanding performance before 2007 but dropped to negative in recent years.

Among the three models, the one that performed the best in recent years was 5MA+Vol; it generated a Sharpe

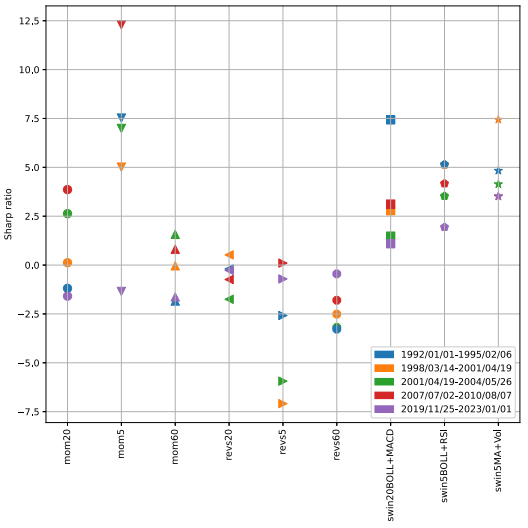


Fig. 6. Scatters of Sharpe ratios for different strategies.

ratio of 3.53, which is significant at the 1% significance level during the 2019–2023 period.

4.2.2. Comparison with traditional technical strategies

In our models, we mainly utilize four technical indicators: the moving average, Bollinger bands, the RSI, and the MACD. Therefore, we also propose four trading rules based on these indicators as benchmarks:

(1). Moving average strategy: Long the asset when its close price is higher than the five-day moving average. Short the asset when its close price is lower than the five-day moving average.
(2). Bollinger bands strategy: Long the asset when its close price is higher than the middle band and not higher than the upper band. Short the asset when its close price is below the middle band and higher than the lower band.
(3). RSI strategy: Long the asset when its RSI(5) is higher than the RSI(10). Short the asset when its RSI(5) is lower than the RSI(10).
(4). MACD strategy: Long the asset when its MACD curve is higher than the signal curve. Short the asset when its MACD curve is lower than the signal curve.

To build portfolios based on technical signals, we evenly allocate our funds across all assets that have triggered valid long or short signals on a given day.

The results in Table 5 show that our models outperform all four trading rules on all test periods.

4.2.3. Comparison with industry portfolios

Our analysis utilizes the daily returns of 10 industry portfolios sourced from Kenneth R. French's comprehensive data repository. Specifically, we employ the regression analyses of Eq. (10) to assess the performance and correlation of the models with the industry portfolios. The constant term α captures the excess return of our portfolios that is not explained by 10 industry portfolios.

$$\begin{aligned} E(R_p) = \alpha &+ \beta_1 R_{\text{NoDur}} + \beta_2 R_{\text{Durbl}} + \beta_3 R_{\text{Manuf}} + \beta_4 R_{\text{Enrgy}} \\ &+ \beta_5 R_{\text{HiTech}} + \beta_6 R_{\text{Telcm}} + \beta_7 R_{\text{Shops}} + \beta_8 R_{\text{Hlth}} \\ &+ \beta_9 R_{\text{Utils}} + \beta_{10} R_{\text{Other}} \end{aligned} \tag{10}$$

As depicted in Table 6, the daily returns of the three ML-based portfolios exhibit negligible correlations with the returns of most of the 10 industry portfolios. Exceptionally, the 5BOLL+RSI model demonstrates a positive correlation (significant at the 5% level) with both the telecom industry and the utilities industry. This phenomenon can be viewed in two ways: (1) the 5BOLL+RSI model prefers assets in those sectors; and (2) the price patterns in these two sectors are more predictable for the specific model. Furthermore, our portfolios generate statistically significant daily alphas, which, when annualized, amount to 5.13%, 7.79%, and 10.51% for each respective portfolio.

4.2.4. Performance analysis using Fama–French five-factor model

Fama and French (2015) proposed a five-factor model that extends the three-factor model by adding profitability and investment factors to market risk, size, and value.

It aims to explain asset returns with these five elements: market premium (RMRF), size (SMB), value (HML), profitability (RMW), and investment (CMA). We utilize this model not only to analyze stylish behaviors of our model but also to detect anomalies in their performance.

It is important to observe that, as indicated in Table 7, all models yield statistically significant positive return anomalies at the 1% significance level. This finding provides additional evidence that our models are capable of identifying previously unexploited profitable patterns within the market. Furthermore, all models exhibit negligible correlations with all factors. Nevertheless, our models demonstrate a significant negative coefficient with respect to the RMW factor, with two models reaching statistical significance at the 5% level and one at the 10% level. This negative coefficient suggests that our models possess a preference for under-profit firms, implying that the pricing patterns of unprofitable firms might be more predictable for models.

4.3. Transaction costs

To assess the economic feasibility of ML portfolios, it is crucial to conduct an analysis concerning models' sensitivity to transaction costs under varied market environments. We first employ the VIX index data from the CBOE website as an indicator of market volatility. Fig. 7 demonstrates the trend of the yearly mean VIX index across past decades. Then we discern volatility periods based on the yearly mean level of the VIX index to four different states: low ($\leq$ 15), moderate ($\leq$ 20 and $>$ 25), high ($\leq$ 25 and $>$ 20), and extreme ($>$ 25). We calculate break-even transaction costs separately on different volatility periods.

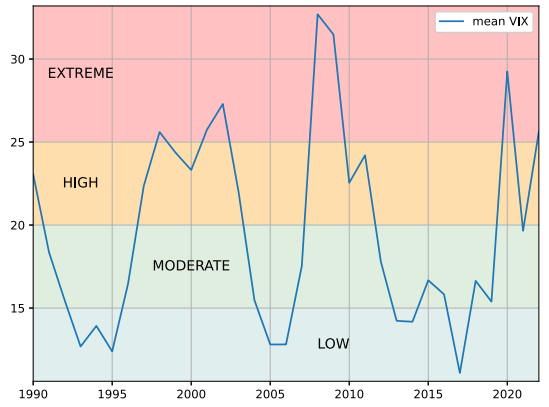
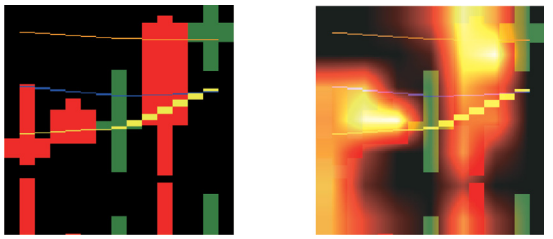


Fig. 7. Yearly mean VIX index from 1990 to 2022.



(a) Original input chart (b) Input chart overlaid with the attention map

Fig. 8. A successfully predicted image of 5MA+Vol (predicted label: 1).

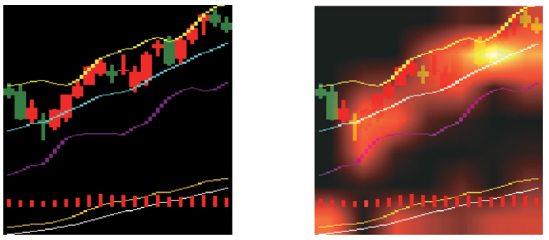
Referring to Table 3, it is pertinent to observe that during periods of low market volatility, three models demonstrate considerable tolerance to transaction costs. A detailed examination reveals that the performance varies across different models. Specifically, the 20BOLL+MACD and 5BOLL+RSI models exhibit a high degree of acceptance of transaction costs during both extreme and low volatility periods. However, these models appear to be more susceptible to increased costs during periods of high and moderate market volatility. In contrast, the 5MA+Vol model's sole area of vulnerability is confined to the moderate volatility period.

4.4. Visualization of attention maps

Our approach to visualizing the attention map of the models is based on the GradCAM technique proposed by Gildenblat and contributors (2021).

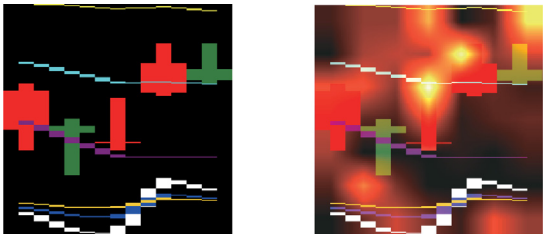
Fig. 8 illustrates an example of attention map visualization. In this instance, there are plural breakthroughs of positive OHLC bars on days 1, 2, and 4 relative to multiple moving averages, which is considered a typical buy signal under technical analysis.

In Fig. 9, the majority of the OHLC bars sit above the midline of the Bollinger bands, while the MACD histogram is predominantly populated with positive values. These indicators collectively convey bullish signals according to the principles of technical analysis.



(a) Original input chart (b) Input chart overlaid with the attention map

Fig. 9. A successfully predicted image of 20BOLL+MACD (predicted label: 1).



(a) Original input chart (b) Input chart overlaid with the attention map

Fig. 10. A successfully predicted image of 5BOLL+RSI (predicted label: 1).

In the case of Fig. 10, the OHLC bar on day 4 crosses the middle band of the Bollinger bands, which is recognized as a buy signal.

Generally, the highlighted regions within these images indicate that all models have effectively identified information provided by technical indicators. It is evident that the models have learned how to utilize technical indicators for recognizing price trends and making decisions based on image patterns.

5. Conclusions

We presented a novel technical analysis approach that leverages visualizations of OHLC charts and various technical indicators, expanding the research agenda proposed by Jiang et al. (2023). Our framework provides a reliable method of visualizing OHLC data and technical indicators by utilizing cutting-edge computer vision models to process image inputs. We shuffled training, validation, and test samples to prevent data snooping and thoroughly evaluated the performance of portfolios based on our model's predictions. To establish benchmarks, we employed multiple momentum and reversal strategies, traditional technical trading rules, industrial portfolios, and the Fama–French five-factor model. Furthermore, a transaction costs sensitivity analysis was conducted to investigate the robustness of the model under various market volatility conditions. Finally, we utilized the GradCAM approach to interpret our model's predictions by visualizing the attention map.

Table 5
Portfolio performance comparison with traditional technical strategies. Specifically, we use high-minus-low portfolios of our models as comparisons.

Periods	swin20BOLL+MACD		swin5BOLL+RSI		swin5MA+Vol		MA		BOLL		MACD		RSI	
	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR	ret	SR
1992/01/01-1995/02/06	***8.45%	***4.18	***14.39%	***3.82	***12.92%	***3.44	−98.46%	−1.68	−98.25%	−1.60	−93.24%	−1.73	−98.25%	−1.70
1998/03/14-2001/04/19	***8.87%	1.16	***11.12%	***2.74	***17.70%	***5.26	−59.83%	−7.39	−53.72%	−6.61	−31.75%	−4.44	−57.74%	−7.78
2001/04/19-2004/05/26	2.85%	0.63	***5.56%	**2.49	***8.15%	***3.31	−50.22%	−7.22	−48.55%	−6.85	−28.99%	−4.32	−50.02%	−7.39
2007/07/02-2010/08/07	***8.08%	***2.67	***8.49%	***3.60	***8.39%	***3.02	−41.82%	−2.43	−36.74%	−2.10	−16.77%	−1.00	−40.67%	−2.43
2019/11/25-2023/01/01	2.80%	0.76	*3.89%	1.52	***10.37%	***3.24	−8.23%	−0.56	−10.89%	−0.68	−3.33%	−0.23	−12.19%	−0.92

Table 6
Regression results of three machine-learning portfolios against 10 industry portfolios. Specifically, we use high-minus-low portfolios of our models as comparisons.

Model	α	NoDur	Durbl	Manuf	Enrgy	HiTec	Telcm	Shops	Hlth	Utils	Other	R^2
swin20BOLL+MACD	***0.0002	0.0014	0.0004	0.0054	−0.0011	−0.001	0.0024	*-0.0060	0.0017	0.0026	−0.0038	0.002
swin5BOLL+RSI	***0.0003	−0.0046	−0.0012	0.003	−0.001	−0.0014	**0.0063	−0.0028	0.0013	**0.0049	−0.0036	0.004
swin5MA+Vol	***0.0004	−0.0067	−0.0007	−0.0037	0.0019	−0.003	0.003	0.0007	0.0041	0.0046	0.0048	0.005

Table 7
Regression results of three machine-learning portfolios against the Fama–French five-factor model. Specifically, we use high-minus-low portfolios of our models as comparisons.

Model	α	Mkt-RF	SMB	HML	RMW	CMA	R^2
swin20BOLL+MACD	***0.0001	−0.0017	0.0035	0.0009	**-0.0100	0.0002	0.002
swin5BOLL+RSI	***0.0002	−0.0007	0.0002	0.001	**−0.0080	*0.0096	0.002
swin5MA+Vol	***0.0003	**0.0047	−0.0035	0.0007	*−0.0086	**0.0126	0.004

Our findings demonstrate several key points. First, our approach shows that it is possible to create portfolios based on a model that takes various visualized technical indicators as inputs. Second, we emphasize the importance of having a high-quality training set, as it can significantly impact the effectiveness of the trained model. Third, we observed that strategies relying solely on historical price trend information tended to perform worse in recent years than in earlier years. The best-performing model was based on technical indicators of multiple moving averages and volume. Last, the visualization of attention maps confirmed that our models can learn from price charts and extract valuable information from them.

Several unresolved limitations remain. First, the mechanism behind toxic samples requires further analysis: How training samples affect the model's performance and how to find the most efficient samples should be carefully investigated. Second, we only tested six technical indicators, but the universe of technical indicators is much wider. Thus, it would be intriguing to examine more indicators with this approach. Third, it is possible to build a model that takes both technical indicators and companies' financial statements as inputs to make more comprehensive predictions in future studies. Fourth, the threshold setting in our validation process did not account for the asymmetry of market returns. A model training framework that addresses this property could be discussed in future studies.

Given artificial intelligence's potential to replace human decision making, we believe that developing a ML-based technical trading agent is both possible and feasible. With carefully selected training sets and robust models, it could yield substantial returns for investors seeking a powerful technical analysis model.

Data and code availability

All source code used to produce the figures and replicate the empirical analysis is publicly available at https://github.com/YiRuiDong/VisualizingTechnicalAnalysis. The datasets underlying the study are archived on Zenodo and can be accessed via the DOI https://doi.org/10.5281/zenodo.19416222.

CRediT authorship contribution statement

Ruidong Yi: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiajie Wang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijforecast.2026.06.002.

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